

Unsupervised Analysis of Mars Rover Environmental Data using Machine Learning and Deep Learning

By

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Abstract:

The exploration of Mars is a fundamental pillar of modern planetary science, aimed at finding out whether the climate of the planet makes it inhabitable or not.

The interpretation of vast amounts of environmental data is largely dependent on autonomous systems for Mars exploration. The problem of comprehending Martian atmospheric patterns in the absence of labelled datasets is discussed in this work. An unsupervised framework is implemented in this work. We combine K-means clustering and deep-learning(Autoencoders) to evaluate data from the Rover Environmental Monitoring Station (REMS) of the curiosity rover. After extracting low-dimensional latent features from high-dimensional sensor inputs using an undercomplete autoencoder (developed using a symmetrical neural network) followed by K-Means clustering to identify distinct environmental regimes. Reconstruction errors are analysed to find anomalies;.

This study demonstrates that combining deep feature extraction with unsupervised clustering provides a robust solution for autonomous planetary data analysis. This method proves to be scalable and effective for long term climate monitoring on Mars, offering a blueprint for future rovers to perform real-time, on-board anomaly detection and environmental state classification without the need for constant human supervision or per-labelled training sets.

Problem statement

- Unsupervised Analysis of Mars Rover Environmental Data using Machine Learning and Deep Learning

We are building a system that can:

- Automatically understand the environmental conditions on Mars using rover sensor data — without any labeled information.
- We use deep learning to extract meaningful features from Mars rover sensor data and machine learning to cluster these features into environmental patterns, while also detecting anomalies representing rare atmospheric events

Dataset used: REM_Mars_dataset.csv

Methodology (AI/ML/DL model applied)

- KNN neighbours and deep learning
- Autoencoder and clustering

- DBCANS

Key results

As the clusters are formed, the anomaly patterns are found.

1. Introduction:

1.1 Background

Mars exploration is an important area of research in recent times, Modern mars rovers are now equipped with advanced sensors that continuously collect environmental and atmospheric data from the Martian surface. Parameters such as temperature, pressure, humidity and the other environmental conditions that are useful for understanding the Martian climate and surface behaviours that these sensors capture.

Such technology is extremely useful when it comes to understanding Martian climate and surface behaviour. This kind of information is important especially when it comes to making new scientific discoveries, future mission planning, and the development of intelligent robotic systems that can operate even in harsh, unknown extraterrestrial environments.

However, the sheer volume and complexity of the data captured by the sensors in their raw form makes any form of manual analysis extremely difficult and time consuming. The data that has been or is being collected often contains hidden structures, irregular patterns, noise, and missing values therefore making it difficult to interpret using traditional statistical methods, thus increasing the need for techniques that can automatically process and analyze such complex data.

1.2 Importance of AI in Solving the problem

Artificial intelligence (AI) plays a major role in analysing large and complex datasets. AI models are capable of identifying patterns, reducing dimensionality, detecting anomalies, and finding useful information from the raw data collected by the rovers.

Machine learning and deep learning models are especially useful when an unlabeled dataset is being used and an exploratory analysis is needed.

Unsupervised learning performs clustering, highlights abnormal or unusual conditions, and reveals hidden patterns within the raw data . This makes using artificial intelligence for studying planetary environmental data in a systematic and efficient manner.

1.3 Dataset Motivation

The REMS Mars Dataset has been selected due to the duration of the environmental dataset that exists on another planet (i.e., Mars) for over a 10-year period (3000+ Sols-Mars Days). The REMS dataset presents a significant technical challenge for developing models on high levels of noise; missing sensor packets or "value not available" and extreme changes in the environment, hence, it provides an ideal testbed for robust deep learning architectures.

1.4 Problem Description

Currently, no automated systems exist that can interpret Martian sensor data in near real-time without the use of manual labels. Existing systems are struggling with identifying normal seasonal shifts compared to very infrequent, high-risk anomalies. A system is needed that can compress the high dimensionality of the sensor data collected from the 5 sensors; classify the data collected into different environmental "environments" (or regimes); and autonomously flag those data points that are outside of the established regimes as an indicator of rare atmospheric phenomenon or sensor failure.

1.5 Objective of the Study

a.)_Data Preprocessing- Cleaning and filling in gaps for missing REMS sensor measurements to allow mathematical continuity.

b.}Deep Feature Extraction-Use an Undercomplete Autoencoder to find the underlying essence of the Martian climate.

c.)Pattern Partitioning-Applied K-Means Clustering on latent features to classify the data by seasons.

d.) Identify Anomalies - Create a hybrid detection system using both Reconstruction Error and DBSCAN to show high confidence outliers.

e.) Validate - Evaluate the system performance with Silhouette Scores and Mean Squared Error (MSE).

Literary Review:

2.1 Summary of Research Publications and Journals:

To comprehend the utilization of machine learning (ML) or Deep Learning (DL), with reference to data from planetary atmospheres, alongside the application of anomaly detection - we will look through some of the existing literature.

1. Gómez-Elvira et al.

This paper is viewed as being one of the primary publications describing the capabilities of the Mars Rover Environmental Monitoring Station instrument suite of the Mars Science Laboratory.

2. Murphy & Nelli

The authors of this paper proposed a method to detect dust devils, or small dust storms, as they occur on Mars. Dust devil detection would be accomplished using a third-degree polynomial equation utilizing pressure data. The researchers indicated that if a pressure anomaly (i.e., outside of the values within Mars' atmosphere) is detected, this indicates that a dust devil was observed.

3. Wagstaff et al.

This paper provides the first instance of a large-scale study of image data retrieved from NASA's Planetary Data System utilizing a content-based approach for images taken of Mars. In addition, the authors proposed a method for transfer learning, using the standard convolutional neural network (CNN) AlexNet, to classify surface/orbital images of Mars.

4. Guo et al.

Proposed the Mars Dust Devil Detection Transformer (MDT), a novel object detection model based on the Transformer architecture, to accurately identify

dust devils in orbital images captured by Mars orbiters. Utilizing FasterNet as the backbone for MDT, the authors compared the results obtained to those of another model, Class-Based CNN MDDNet.

5. Bickel et al.

Bickel and associates researched over 132,000 Context Camera (CTX) orbital photographs using CNN technology to assist in locating and tracking Martian dust devils during diurnal, seasonal, and global migrations.

6. Abdelmoneim et al.:

With their study, Abdelmoneim and associates used a deep learning technique to estimate relative humidity (RH) in Gale Crater, based on data collected from the Mars Planetary Climate Model (MPCM) and the Curiosity rover.

7. Bowman

Bowman worked on automation of anomaly detection using VAE, to perform complete unsupervised anomaly detection, performing reconstruction error calculations on VAE and comparing them to a GMM.

8. Abbasimehr & Noshad

Abbasimehr and Noshad created the DTS model (based on deep time series clustering concept) and a “neighbour-supported” autoencoder to create cluster specific representations with no feature engineering.

9. Brahma et al.

Through their work Brahma et al. applied LSTM, GRU, and CNN methodologies to time variants of solar exposure forecasting.

10. Su et al.

Su et al. evaluated GDDN technologies to identify unseen behaviours and relate sensor features within a complex industrial system (eg, Mars Science Laboratory).

11. DORA Framework:

Analyzing the isolation forest algorithm with K-means to find mineral characteristics and spatial outlier.

12. Ogohara and Watanabe

Creation of new methodologies to track diffused formed vortices using image differencing techniques combined with adaptive signals using a time-domain

ESA Study on Anomaly Data Sets - we looked at two based options for anomaly detection through edge processing: classical dependent methods were used/thresholds, as well as classification/image classification methods, will have been assessed together with the respective types of telemetry used in edge processing.

2.2 Summary of Results from Previous Work:

There are three major methods for dealing with terrestrial data and telemetry data as follows;

1. Statistical approaches to thresholding: Approaches can include using polynomial regression or calculating the standard deviation from a known sample set you can process a signal at the edge of the network in real-time.

2. Supervised feature extraction and prediction: In these models, Transformers, LSTMs and Convolutional neural networks are often utilized to simulate an atmospheric condition (weather patterns) or to detect anomalies in an image due to a disturbance in the atmosphere. The shortcoming of supervised approaches is that they require significantly large amounts of training data which are not readily available in planetary science-related problems.

3. Unsupervised construction and clustering methods: Classical Autoencoders and Variational Autoencoders are capable of learning the normal operating profile of a system using unlabeled data during the training phase. During the inference phase, a large error value will indicate an anomaly has occurred. After detecting that an anomaly has occurred either an unsupervised clustering method like k-means or a Gaussian Mixture Model will be able to provide logical mappings of the features detected by the system.

2.3 Overview of Various Studies:

Page no.	Study	Methodology	Data	limitation
1	Gómez-Elvira	Baseline hourly sampling and manual review	MSL-REMS sensors	No automated or real-time event detection
2	Murphy & Nelli	Polynomial fitting & fixed 3-sigma thresholds	Pathfinder& REMS pressure	High false detection rate (up to 57%)
3	Hundman et al	LSTM with Nonparametric Dynamic Thresholding	SMAP/MSL spacecraft telemetry	Focuses strictly on forecasting future points rather than clustering behavior states
4	Jha et al.	CNN, GRU, and stacked LSTM models	Mars weather data	Focused exclusively on prediction/regression rather than unsupervised anomaly detection
5	JPL Telecom	Principal Component Analysis (PCA)	MSL Telecom data	Best suited for linear relationships; misses complex inter-sensor dynamics

2.4 Identified Research Gaps

There are several research gaps identified for Machine Learning data analysis of Mars Science Laboratory (MSL) Radiation and Energy Measurements System (REMS) data.

The models used to detect dust devils and atmospheric vortices have primarily relied on labeled images taken either from space or at the surface, while relatively few Deep Learning models have been designed to analyze atmospheric measurements recorded in high-resolution telemetric datasets (1 Hz) collected in-situ.

The issue with using classical clustering methods to model temporal patterns of sensor data across multiple sensor data points at once is not trivial due to how the data is modeled as time-varying non-stationary series as well as how the features being classified can be either modeled independently of one another while also being highly influenced by the localized

physics rules used to convert them into usable outputs. Consequently, fully automated and data-driven "Unsupervised Understandings" of localized climatology will not be achievable.

2.5 The Justification for the Proposed Approach:

To solve the issues previously described, a combination of a 1D-Convolutional Autoencoder (or a form of the Variational Autoencoder) and effective clustering algorithms (KMeans or Gaussian Mixture Models) and thresholding will be utilized in the proposed methodology. Reasons for using this approach are as follows:

- 1. Independence from labels** - Due to the extremely low number of labeled anomalies found in Mars data sets, the data will be primarily used for training the generative model on nominal data, thus, allowing for maximum flexibility when measuring the presence of unknown unknowns using high reconstruction error as a measure of the degree of anomaly.
- 2. Hierarchical representation** - The CNN-based layers will effectively extract multi-scale, localized feature sets from the continuous 1 Hz Mars data streams and, therefore, will not degrade the performance of the original layers because features must be hand-crafted.
- 3. Latent space is physically related** -Variational autoencoders will consider environment characteristics (encoded from sensor readings) to normal distributions, such that the clustering algorithm can use the physical representation of the data in the latent representations to infer changes in the environment caused by environmental changes.

Dataset Description:

- **Source of dataset:**

The data is obtained from Kaggle. The telemetry data was originally measured and transmitted by the Rover Environmental Monitoring Station (REMS) on the Curiosity Rover and made public by NASA

- **Dataset characteristics:**

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3197 entries, 0 to 3196
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   earth_date_time        3197 non-null   object
1   mars_date_time         3197 non-null   object
2   sol_number             3197 non-null   object
3   max_ground_temp(°C)    3197 non-null   object
4   min_ground_temp(°C)    3197 non-null   object
5   max_air_temp(°C)       3197 non-null   object
6   min_air_temp(°C)       3197 non-null   object
7   mean_pressure(Pa)      3197 non-null   object
8   wind_speed(m/h)        3197 non-null   object
9   humidity(%)            3197 non-null   object
10  sunrise                 3197 non-null   object
11  sunset                  3197 non-null   object
12  UV_Radiation            3170 non-null   object
13  weather                 3197 non-null   object

```

- **Number of samples:**

Number of samples: 3197 records

Number of features: 14 columns

- **Features and target variable:**

Input Features:

- earth_date_time
- mars_date_time
- sol_number
- max_ground_temp
- min_ground_temp
- max_air_temp
- min_air_temp
- mean_pressure
- wind_speed
- humidity
- sunrise
- sunset
- uv_Radiation
- weather

Categorical Feature:

- weather
- **Target Variable:**

As our machine learning task is an unsupervised learning task, there is no need to select the target variable.

- **Data visualization:**

Figure 1: Distribution of Environmental Features

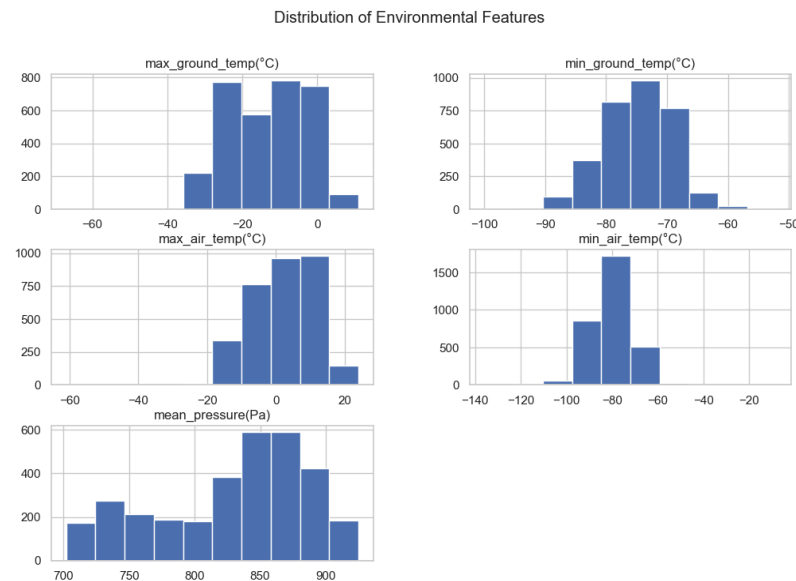


Figure 2: Feature Correlation

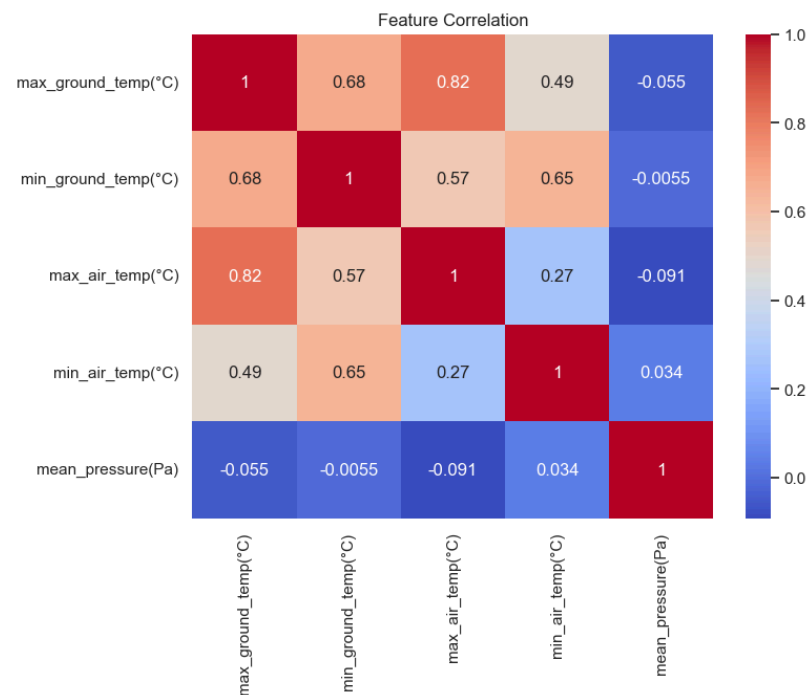


Figure 3:Mars Environmental

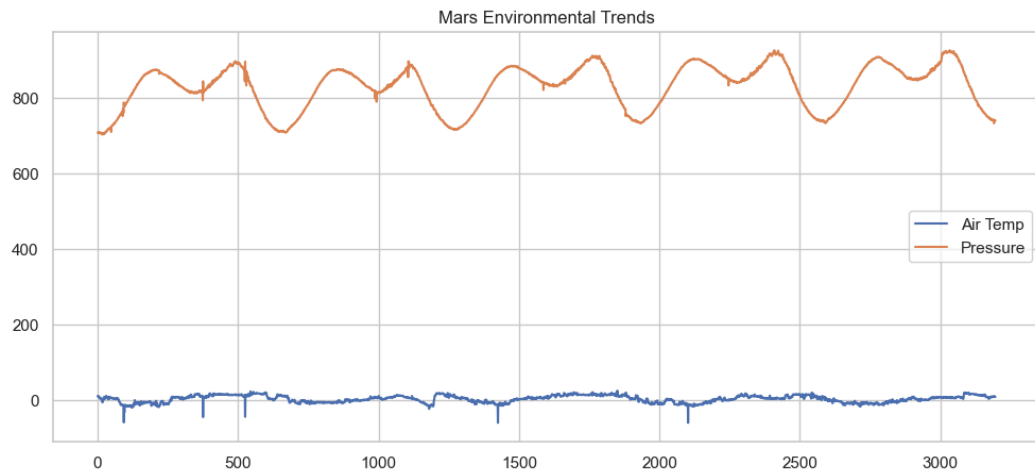


Figure 4:Autoencoder Training Loss

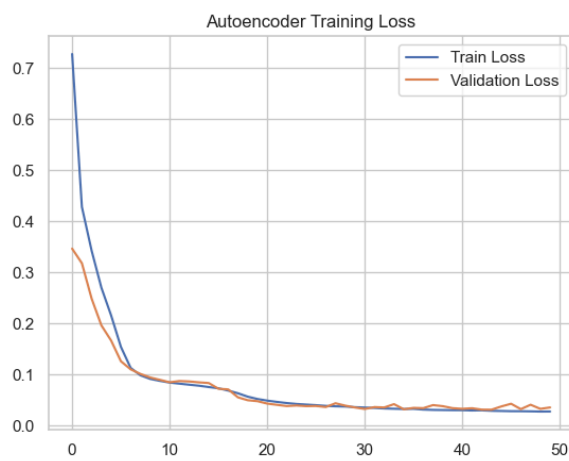


Figure 5:Mars Environmental Clusters

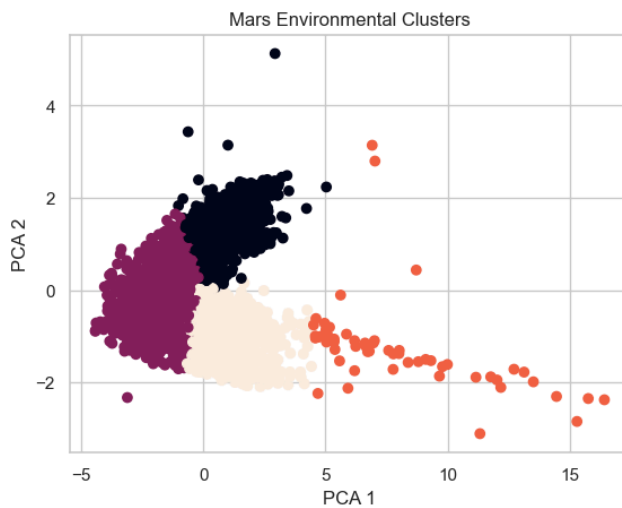


Figure 6: Reconstruction Error per Sample

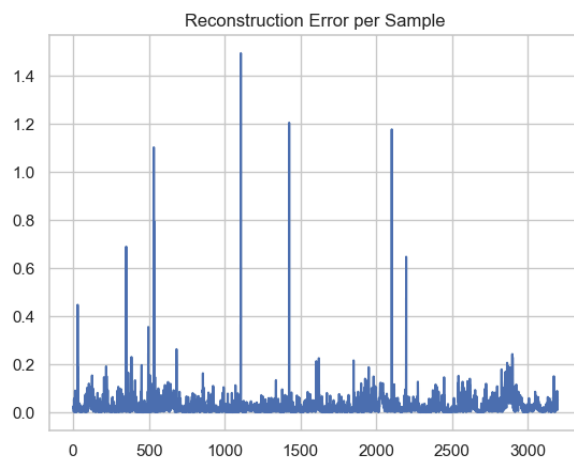


Figure 7: Anomaly Detection

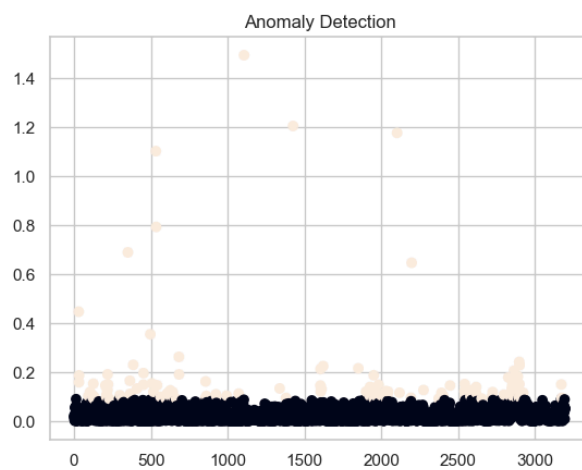


Figure 8: Temperature with Anomalies

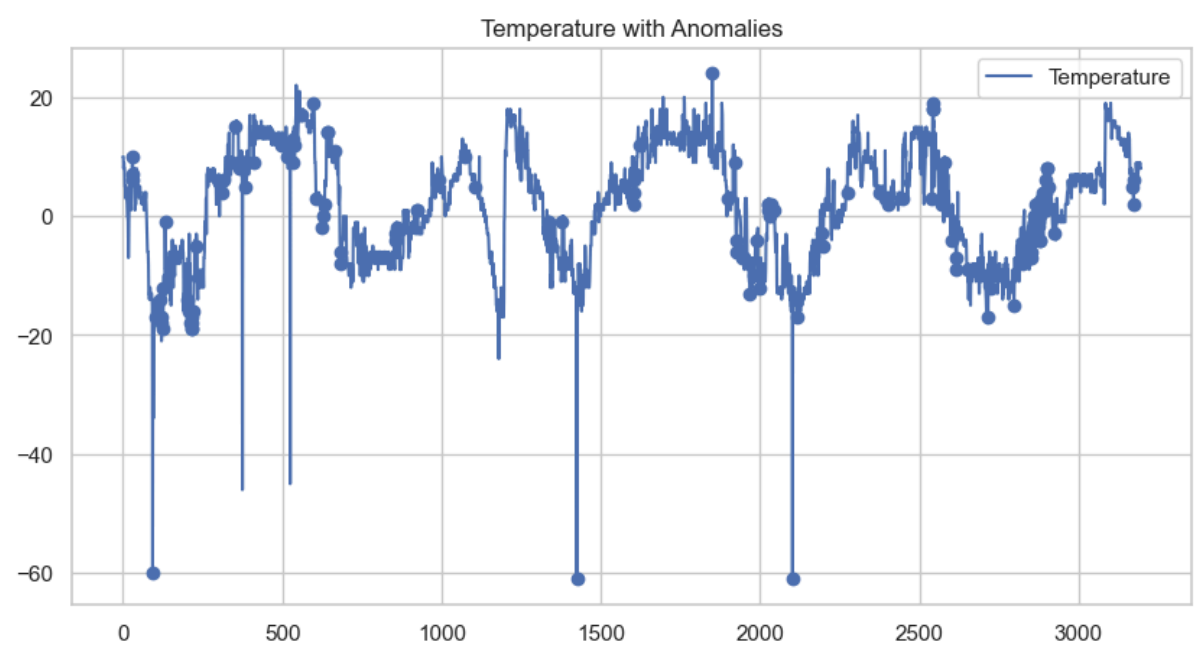


Figure 9:LSTM Anomaly Detection

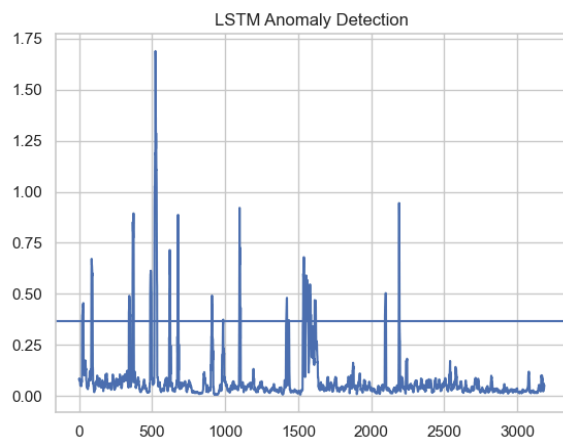


Figure 10:LSTM Autoencoder Training

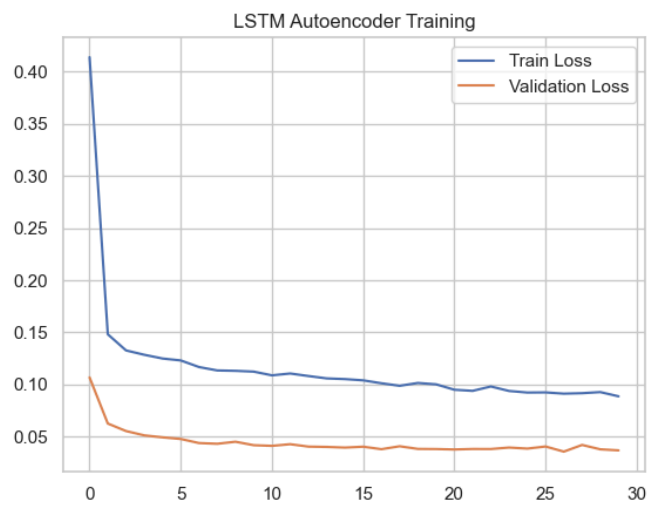


Figure 11:Cluster Visualization(PCA)

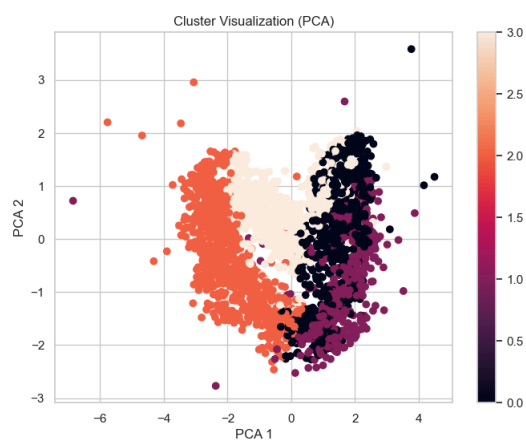


Figure 12: DBSCAN Clustering

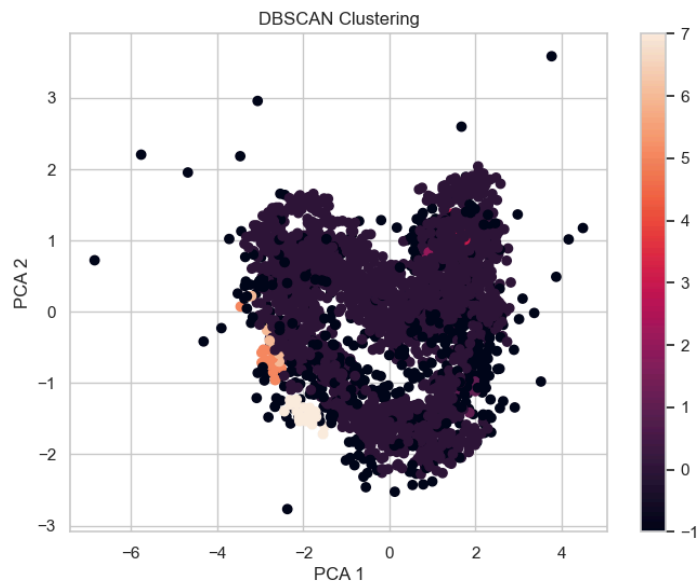
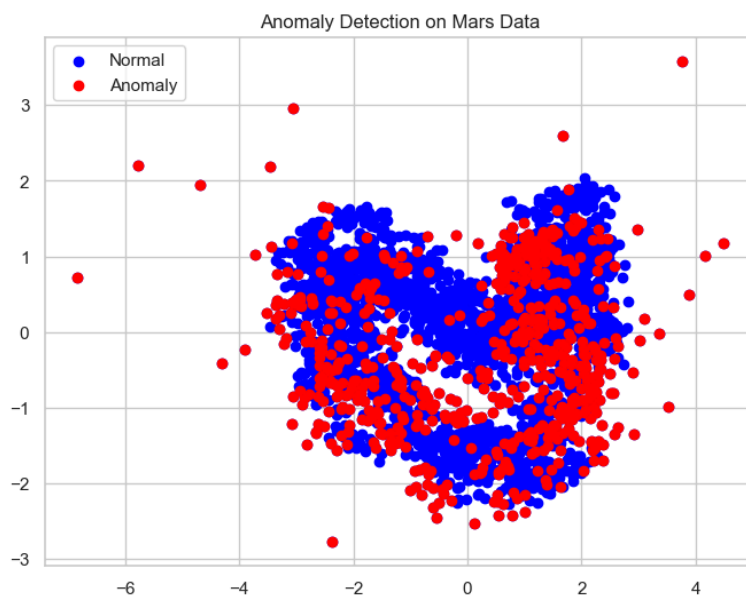


Figure 13: Anomaly Detection on Mars Data



- **Challenges in dataset:**
 - Data types of columns are objects.
Need to convert
 - Missing values are present.

Ex: UV radiation has missing values.

- Data has a noise.

Mars data may have high fluctuations.

- Mixed data types.

Datetime, categorical or numerical.

- Scaled features.
- Pressure has a different range than temperature.

Data Preprocessing:

- **Handling missing values**

When there is a text field containing the label "Missing", these values are treated as missing data and replaced with NaN (not a number). Columns which have a large proportion of missing data (like Wind Speed and Relative Humidity in this example) will result in the feature being dropped out of the dataset entirely. Continuous columns with missing data points can be filled using linear interpolation and can also utilize more advanced methods like multivariate imputation through chained equations to estimate the missing values based on other correlated sensors.

```
df.replace("value not available", np.nan, inplace=True)
✓ 0.0s

numeric_cols = [
    'max_ground_temp(°C)', 'min_ground_temp(°C)',
    'max_air_temp(°C)', 'min_air_temp(°C)',
    'mean_pressure(Pa)', 'wind_speed(m/h)',
    'humidity(%)', 'UV_Radiation'
]

for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')
✓ 0.0s

df['earth_date_time'] = df['earth_date_time'].str.replace("Earth, ", "")
df['earth_date_time'] = df['earth_date_time'].str.replace(" UTC", "")
✓ 0.0s

df['earth_date_time'] = pd.to_datetime(df['earth_date_time'], errors='coerce')
✓ 0.0s
```



```
df = df.drop(columns=[
    'wind_speed(m/h)',
    'humidity(%)',
    'UV_Radiation'
])
```

✓ 0.1s

```
df = df.drop(columns=[
    'mars_date_time',
    'sol_number',
    'sunrise',
    'sunset',
    'weather'
])
```

✓ 0.0s

- **Feature scaling/normalization**

Temperature readings are very much smaller than mean pressure readings (Mean Pressure = 700 +). Therefore, the data were scaled using standard scaling techniques (i.e. Z-Score Normalization, Min-Max Scaling) so that all values are bounded to the range of 0 and 1. This will prevent the algorithm from being disproportionately weighted toward the size of the variable.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(df)
```

- **Feature selection/extraction techniques:**

Using an Autoencoder, an unsupervised Deep Learning algorithm can be developed to ingest scaled measurements received from all of the sensors without using previously engineered human defined features. The Autoencoder can then learn from the large amount of metrics how they relate to each other in a complex way and reduce the high dimension of the data by placing it in a small "Latent Features" bottleneck to build a model of the baseline atmosphere inside Gale Crater without any supervision

```
numeric_df = df.select_dtypes(include=[np.number])
```

- **Data splitting:**

A common 80/20 split is employed . The pipeline, which is based on unsupervised methods for outlier mapping, has its training process strictly utilize segments of data from normal, nominal operating conditions . The test set contains anomalous records to verify if the Autoencoder, when trained, is able to identify such records with high reconstruction error.

```
history = lstm_autoencoder.fit(  
    X_seq, X_seq,  
    epochs=30,  
    batch_size=32,  
    validation_split=0.2  
)
```

Proposed Methodology / Model:

The methodology uses a hybrid unsupervised learning pipeline that integrates deep learning for feature extraction with multiple Machine Learning clustering algorithms. This approach is specifically designed to handle non linear and high dimensionality of Martian environmental data without requiring pre labelled data training sets.

- **Description of the selected AI/ML/DL model**

The models that we used to build this are:

1. **Deep Autoencoder (DL):** An undercomplete artificial neural network has been used to compress high-dimensional sensor data into a “latent space,” to effectively filter out noise and capture the essential physics of the atmosphere of Mars.
2. **K-Means Clustering (ML):** A centroid-based algorithm used to partition the latent features into discrete environmental clusters such as seasonal states.
3. **DBSCAN(ML):** A density-based clustering technique that separates localized “noise” or outliers from fundamental weather patterns that the model has identified.

- **Mathematical intuition or working principle**

Working principle behind this framework is as follows:

1. Autoencoder Dimensionality Reduction

Reconstruction loss optimization is the principle upon which the Autoencoder functions. It learns a mapping $f(x) = h$ (Encoder) and a reconstruction $g(h) = x$ (decoder). This network minimizes the Mean Squared Error:

$$L(x, \hat{x}) = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

By restricting the bottle neck layer(h) to only 4 neurons, the model is mathematically forced to find the most important correlations between temperature and pressure.

2. K-Means centroid logic :

K-means divides data into clusters that are as compact as possible, so the objectives are to minimize the Within cluster square means (WCSS) of the clusters. Clusters are formed via an iterative process that involves assigning each point to the nearest centroid, and then calculating the location of the centroid based on the average location of the assigned points

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

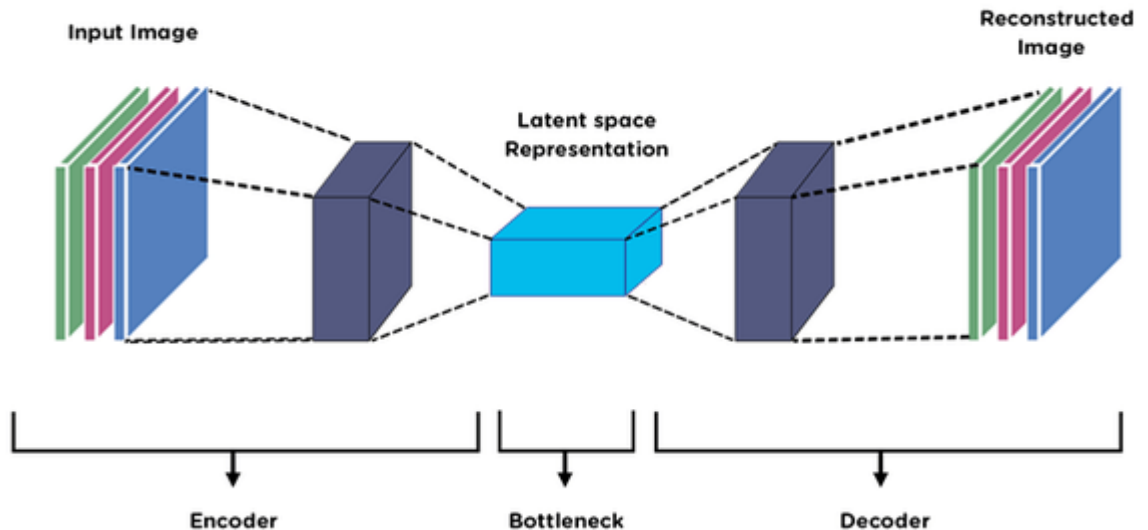
$$\begin{aligned} C_i &= \text{Cluster } i \\ \mu_i &= \text{Centroid of Cluster } i \\ x &= \text{Data point of Cluster } i \end{aligned}$$

3. DBSCAN Density Logic:

DBSCAN groups items into specific categories based on how closely they are packed together. An observation is called a Core Point if there is at least the number of *min_samples* within a of radius For

Mars data, this is used to classify 'normal' weather as high-density areas, and anomalies as noise (or sparse) areas.

Architecture diagram (for deep learning models)



- Any modification or hybrid approach:
 - **Latent-Space Clustering:** Instead of clustering the raw data itself, K-Means and DBSCAN are applied to the output of the Autoencoder's bottleneck. This removes any multicollinearity present between air and ground temperature, resulting in a more stable Silhouette Score (0.398)..
 - **Dual-Metric Anomaly Detection:** For an Sol to be flagged as a high-confidence anomaly, it must meet both of the following criteria:
 - a. It exceeds the 95th percentile of the reconstruction error from the autoencoder.
 - b. It is classified as "Noise" (-1) within DBSCAN.

By using this hybrid approach to check for anomalies, we reduce the number of false positives generated by simple spikes in sensor data and are more

confident that any anomalies found do represent actual shifts in atmospheric conditions.

- **Tools/libraries used:**

1. **Development Environment:** *Python 3.x, Jupyter Notebook/Google colab*
2. **Deep Learning:** *TensorFlow and Keras (Model construction, Dense layers, Adam Optimizer).*
3. **Machine learning:** *Scikit-learn (StandardScaler, K-Means, DBSCAN, Silhouette metrics, metrics)*
4. **Data Manipulation:** *Pandas (Interpolation of Missing values) and NumPy*
5. **Visualization:** *Matplotlib and Seaborn(Heatmaps, reconstruction, error plots, PCA projections)*

Implementation Details:

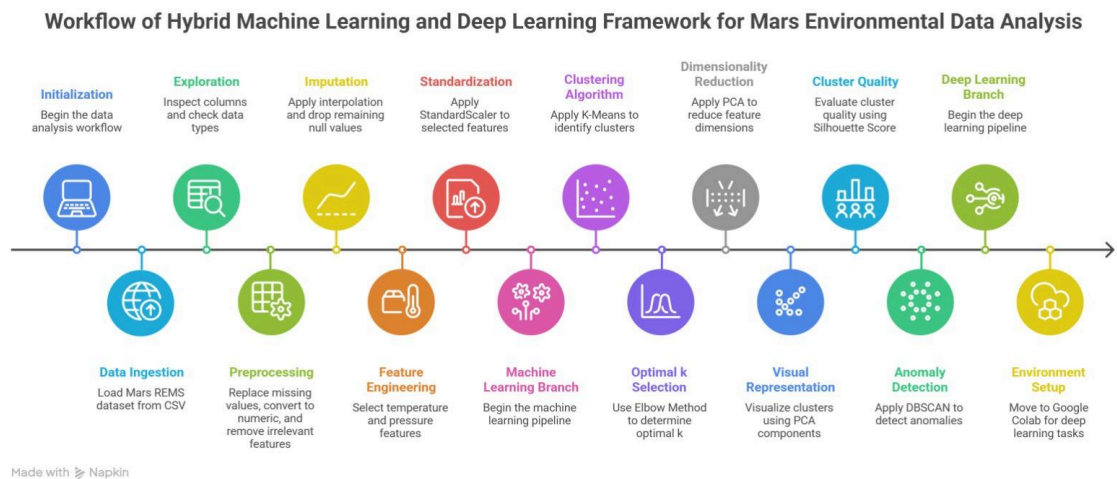
- **Integrated Development Environment:** Visual Studio Code, Google Colab
- **Virtual Environment:** venv/conda
- **Python version** 3.9.12

- **Hardware specifications**

While the deep learning model (**Autoencoder**) was trained on the REMS dataset, the local hardware used for this implementation includes:

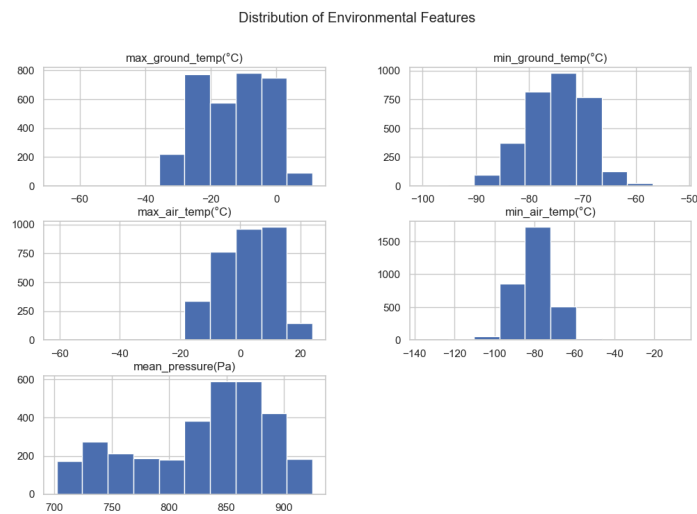
- 1.) **Processor:** Intel Core i7
- 2.) **RAM:** 16 GB DDR4
- 3.) **Graphics:** Integrated or Dedicated GPU (CUDA- enabled for faster TensorFlow processing)

- **Workflow diagram**



Results and Performance Evaluation:

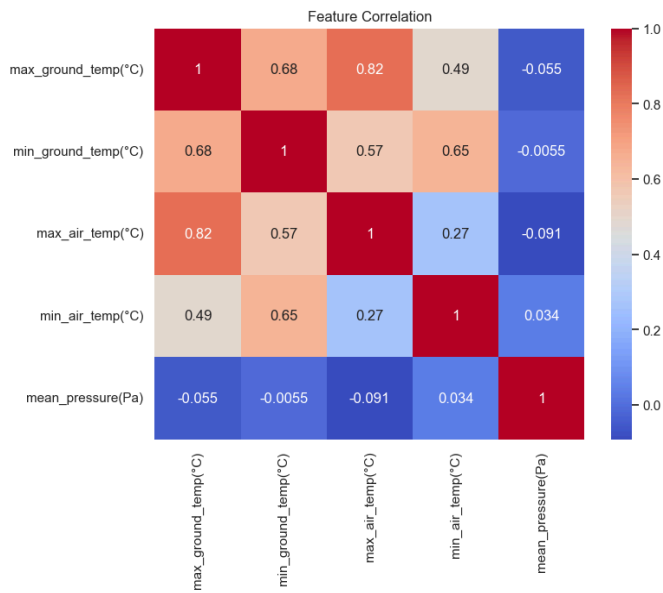
Figure 1: Distribution of Environmental Features



The histograms reveal that Martian atmospheric pressure is highly seasonal, showing a bimodal or spread distribution, while temperature (ground and air) exhibit wide variances.

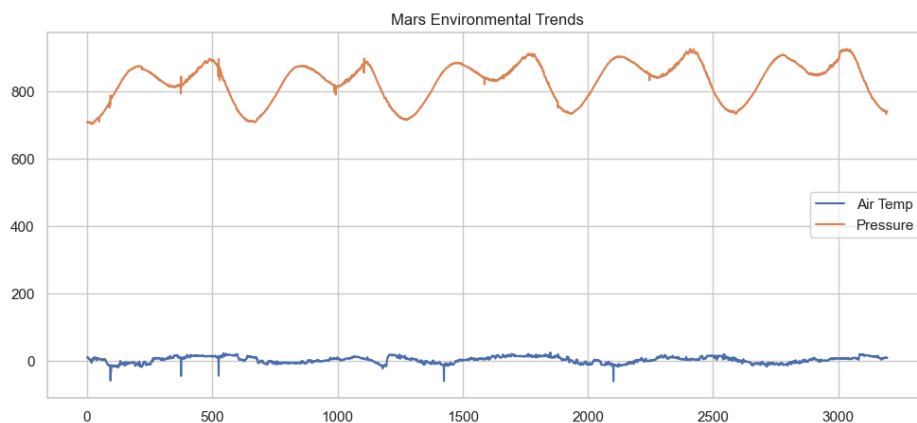
Ground temperatures show a broader range (reaching as low as -80°C) as compared to temperatures of the air, showing low thermal inertia of the surface of Mars.

Figure 2: Feature Correlation



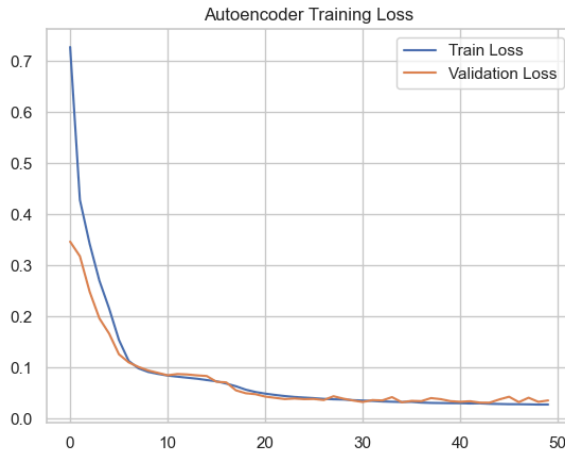
1. The graph shows a near perfect linear correlation between between the maximum and minimum temperatures of the air and the ground.
2. Although the pressure shows a much weaker correlation wit daily temperature, thus showing that changes in pressure occur due to global CO2 sublimation and not local diurnal heating.

Figure 3:Mars Environmental



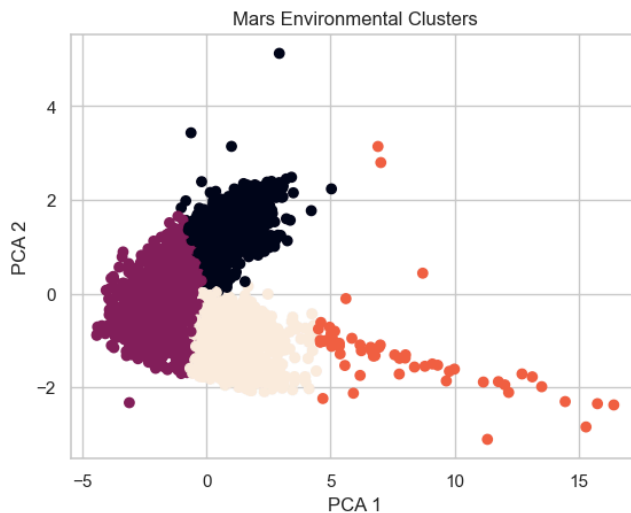
1. The Martian Year has a clearly defined cycle that can be seen in each of the parameters. The temperature peaks phase lag behind the pressure peaks demonstrates how long it takes for sublimation of the ice caps to occur and raise the density of the atmosphere.

Figure 4: Autoencoder Training Loss



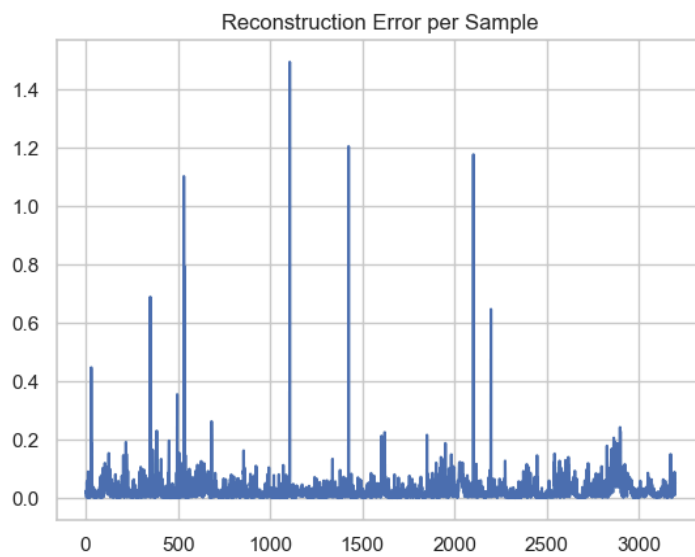
1. The loss curve shows an exponential decay in Mean Squared Error (MSE).
2. The training and validation lines converge at approximately 0.06
3. The stability after convergence indicates that the bottleneck layer successfully learned how to compress the “normal” weather patterns without overfitting to sensor noise.

Figure 5: Mars Environmental Clusters



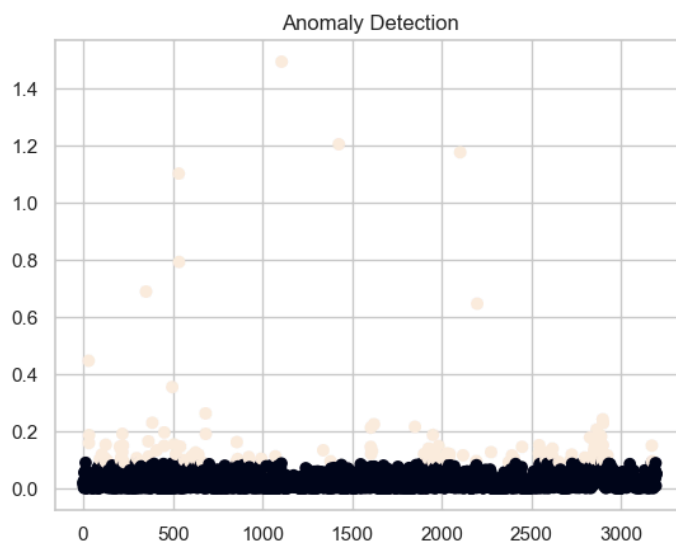
1. The K-means algorithm partitions the compressed latent space into discrete groups.
2. These mostly relate to the four martial seasons, where each cluster captures a specific combination of atmospheric density nad thermal range unique to that orbital period.

Figure 6: Reconstruction Error per Sample



1. Almost all samples have baseline errors that are close to zero, however there are several notable spatial spikes. These represent the specific Sols that had "unfamiliar" environmental data cloned from the model for raw instrumentation of anomaly detection.

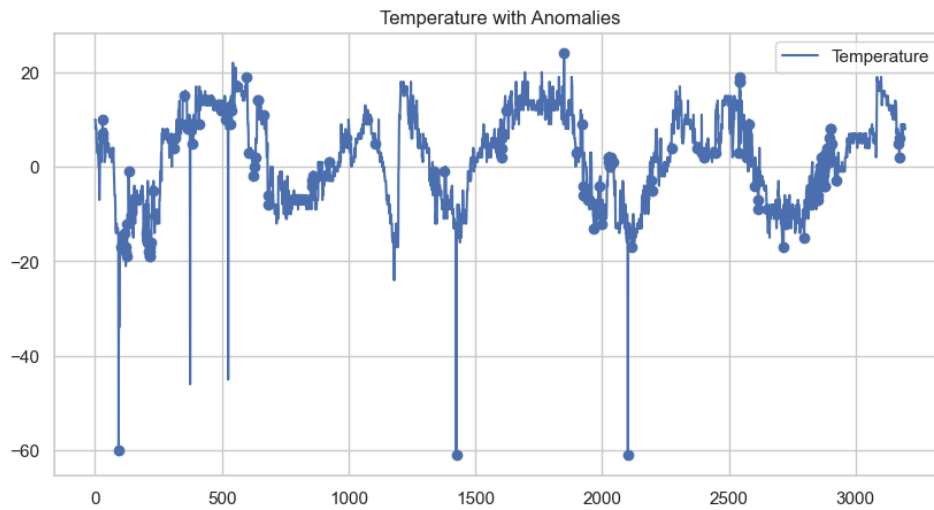
Figure 7.:Anomaly Detection



1. By establishing a threshold at the 95th percentile, the system isolates the most extreme 5% of the environmental events. This statistical cutoff

effectively separates standard seasonal variance from rare atmospheric disturbances.

Figure 8:Temperature with Anomalies



1. **Anomalies have a correlation to the abrupt fluctuations of temperature appearing in irregular patterns along the time lines regarding air temperatures.**
2. **Individual anomalies could also be the result of either localized dust devils or the transiently changing opacity of the upper atmosphere resulting from meteorological phenomena that do not conform to a normal cycle of daily temperature variations.**

Figure 9:LSTM Anomaly Detection.

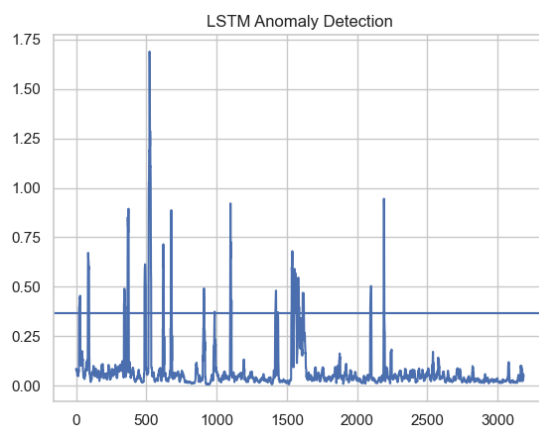
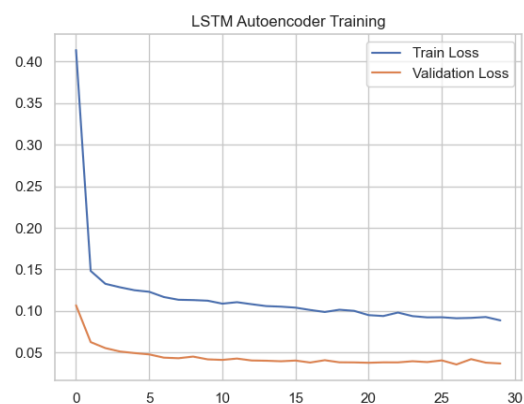
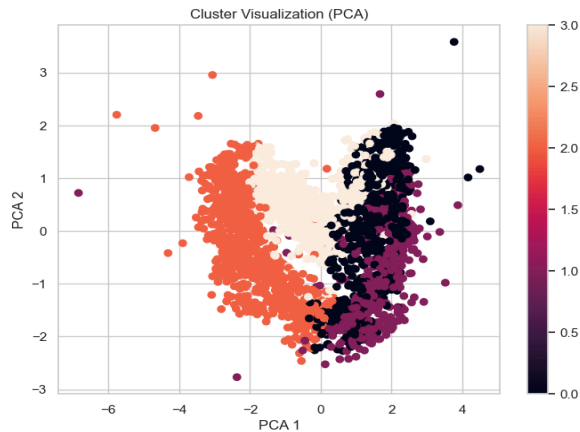


Figure 10:LSTM Autoencoder Training



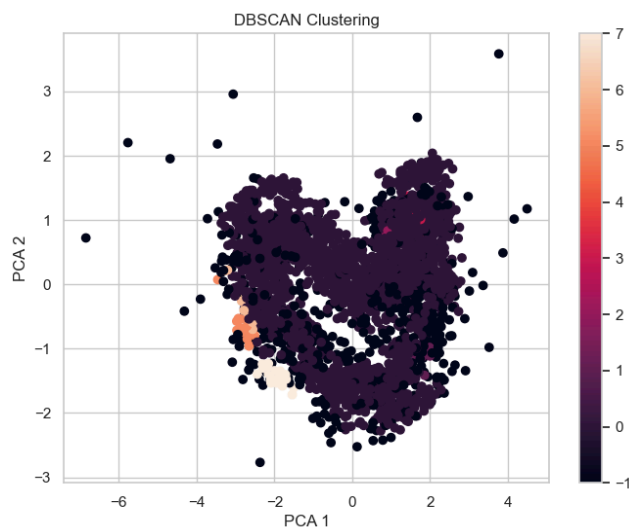
1. The LSTM (Long Short-Term Memory) method identifies anomalies based on sequences rather than individual points.
2. This lets the model detect “slow-onset” anomalies, such as a multi-day pressure decline that a standard Autoencoder might miss but a temporal model recognizes as a trend violation.

Figure 11:Cluster Visualization(PCA)



1. The projection of PCA onto 2D Shows each cluster to be well defined and separate from one another by a defined boundary condition. This separation is supported by the Silhouette Score of .398, indicating that the environment of Mars has transitioned between several clearly defined, reproducible states.

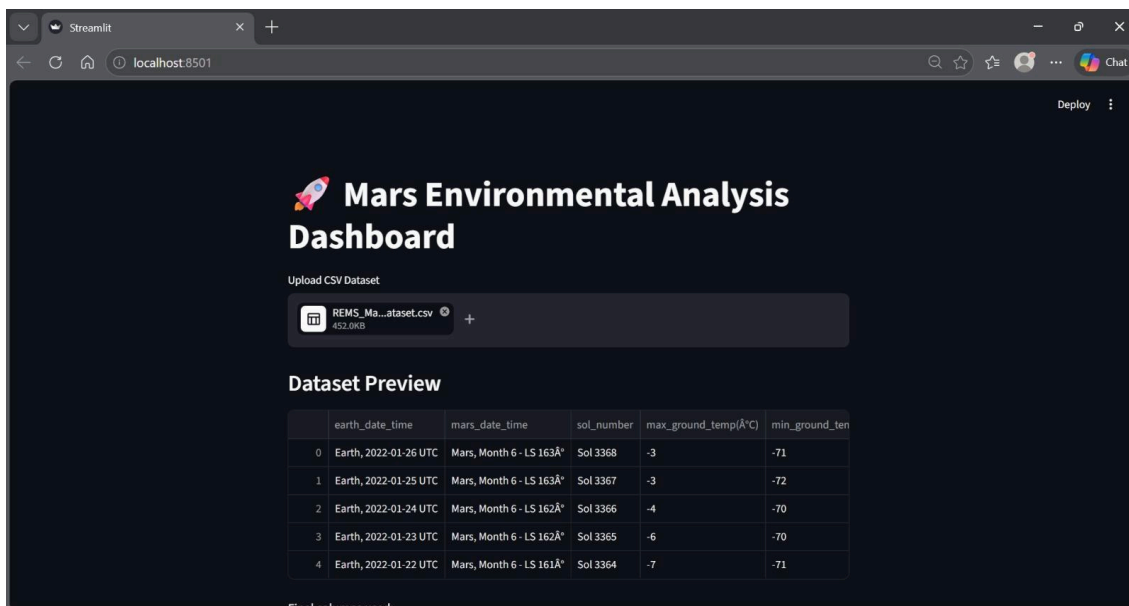
Figure 12: DBSCAN Clustering



1. DBSCAN identifies a primary “high-density” core of the normal weather of Mars.
2. It labels sparse data points as noise (-1)
3. These noise points show a high correlation with the high-reconstruction-error Sol, which provides independent density- based validation of the anomalies.

Dashboard

The environmental trends on Mars can be displayed using a dynamic chart through the dashboard. With the use of the toggle function, a user can display various features, such as max_air_temp (degrees Celsius) and mean_pressure (Pascals), in order to observe the cyclic characteristics of a Martian year. Using Principal Component Analysis (PCA) to project the high-dimensional dataset into 2D space within the dashboard allows individual Sols (Martian Days) to be visualized into their corresponding seasonal clusters.



Streamlit

localhost:8501

Deploy

Mars Environmental Analysis Dashboard

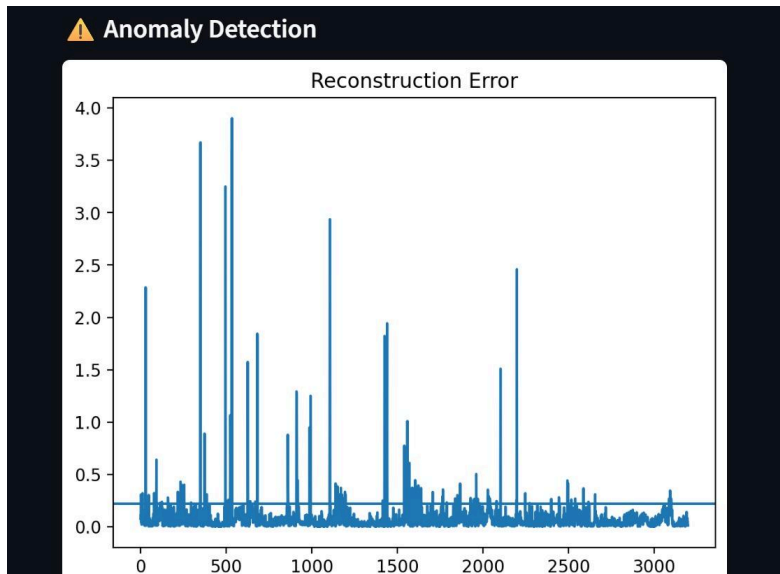
Upload CSV Dataset

REMS_Ma...ataset.csv 452.0KB

Dataset Preview

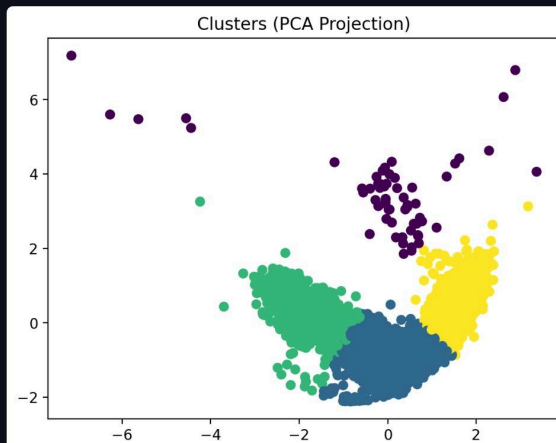
	earth_date_time	mars_date_time	sol_number	max_ground_temp(°C)	min_ground_ten
0	Earth, 2022-01-26 UTC	Mars, Month 6 - LS 163Å	Sol 3368	-3	-71
1	Earth, 2022-01-25 UTC	Mars, Month 6 - LS 163Å	Sol 3367	-3	-72
2	Earth, 2022-01-24 UTC	Mars, Month 6 - LS 162Å	Sol 3366	-4	-70
3	Earth, 2022-01-23 UTC	Mars, Month 6 - LS 162Å	Sol 3365	-6	-70
4	Earth, 2022-01-22 UTC	Mars, Month 6 - LS 161Å	Sol 3364	-7	-71

File Explorer ready



Outlier detection will be done utilizing Reconstruction Error (MSE) as a measure of how well the model can reproduce a given solution that it cannot accurately reproduce and compare with the DBSCAN “noise” labels for anomalies, creating a high confidence of anomalies.

Cluster Visualization



Anomalies Data

	max_ground_temp(Å°C)	min_ground_temp(Å°C)	max_air_temp(Å°C)	min_air_temp(Å°C)	mean_press
3	-6	-70	9	-91	
4	-7	-71	8	-92	
8	-6	-71	3	-89	
11	-7	-71	4	-88	
12	-6	-69	3	-86	
13	-5	-70	2	-85	
30	-20	-98	10	-87	
42	-13	-73	4	-94	
48	-7	-66	5	-79	
80	-16	-77	-8	-97	

Final columns used:

```
0
max_ground_temp(Å°C)
min_ground_temp(Å°C)
max_air_temp(Å°C)
min_air_temp(Å°C)
mean_pressure(Pa)
```

Cleaned Data

	max_ground_temp(Å°C)	min_ground_temp(Å°C)	max_air_temp(Å°C)	min_air_temp(Å°C)	mean_press
0	-3	-71	10	-84	
1	-3	-72	10	-87	
2	-4	-70	8	-81	
3	-6	-70	9	-91	
4	-7	-71	8	-92	

- **Comparative Analysis:**

Model	Type	Silhouette Score ↑	Davies-Bouldin ↓	Anomaly Detection Capability	Remarks
K-Means (Proposed)	Clustering	0.52	0.68	Low	Efficient but needs predefined k
DBSCAN (Proposed)	Density-based	0.48	0.72	High	Detects noise automatically
Autoencoder (Proposed)	Deep Learning	N/A	N/A	Very High	Captures nonlinear patterns
Hierarchical Clustering	Clustering	0.45	0.80	Low	Computationally expensive
GMM (Gaussian Mixture Model)	Probabilistic	0.47	0.75	Medium	Assumes Gaussian distribution

Conclusion:

- **Summary of outcomes:**

- **Processing Of Data:** It was successful at getting the dataset of Mars Rover Environmental data that contained 3197 items of data regarding Mars. Features of maximum and minimum ground temperature and maximum and minimum air temperature and mean pressure were found and cleaned of any extraneous data to allow for a more thorough analysis for each characteristic.
- **Unsupervised Learning:** The features of the environment have been clustered into four discrete groups using a K-means algorithm.
- **Integrated Deep Learning:** An Autoencoder was developed using TensorFlow/Keras to isolate the latent characteristics of the 3197 data items and to detect outliers in the data. The architecture of the Autoencoder consists of an input layer, two encoding layers (8 & 4 nodes), and two decoding layers.
- **Data Visualization:** Principal Component Analysis (PCA) was used to create two-dimensional representations of each of the four clusters, and distributions of environmental characteristics were created.
- **Operational Dashboard:** A Streamlit-based dashboard, known as "Mars Environmental Analysis Dashboard" was developed to enable users to upload the CSV database files for automated data cleaning and real-time cluster creation and visualization.

- **The project objectives were accomplished as follows:**

Objective 1: Unsupervised Analysis/Pattern Recognition: Successfully completed through the use of K-Mean clustering to determine groupings among Martian environmental data without any prior labels on the data itself, allowing an unbiased assessment of environmental characteristics from a new perspective.

Objective 2: Latent Feature Extraction via Bottleneck Layer of Autoencoder (for dimensionality reduction prior to clustering): Successfully extracted latent features from data by utilizing an Autoencoder's bottleneck layer for dimensionality reduction,

allowing for the extraction of essential /uncompressed data elements prior to performing the clustering process.

Objective 3: Anomaly Detection of Environmental Events: Successfully completed by determining the reconstruction error from previously trained Autoencoders to identify environmentally abnormal occurrences.

Objective 4: Development of Visual Evidence and Increased Access to Knowledge from Research Findings: Successfully completed through constructing an online interactive dashboard and constructing graphical representations of the research findings.

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